

In-Class Assignment: Perceptron and Multilayer Perceptron

Please answer the following questions.

Question 1 and 2 : please refer to the [Copy of Perceptron.ipynb](#)

Question 3 and 4 : please refer to the [Copy of 1 - Multilayer Perceptron.ipynb](#)

Question 1: Convergence Condition

What is the essential condition regarding the dataset for the Perceptron algorithm to be guaranteed to converge?

Question 2: Handling Linearly Non-Separable Data

What happens to the learning process if the data is not linearly separable?

Also, what measure is taken in the code of this notebook to prevent an infinite loop?

Question 3: Training Process

Outline the key differences between the train and evaluate functions regarding gradient calculations.

Question 4: Visualization Tools

What insights do PCA and t-SNE provide regarding the model's learning?